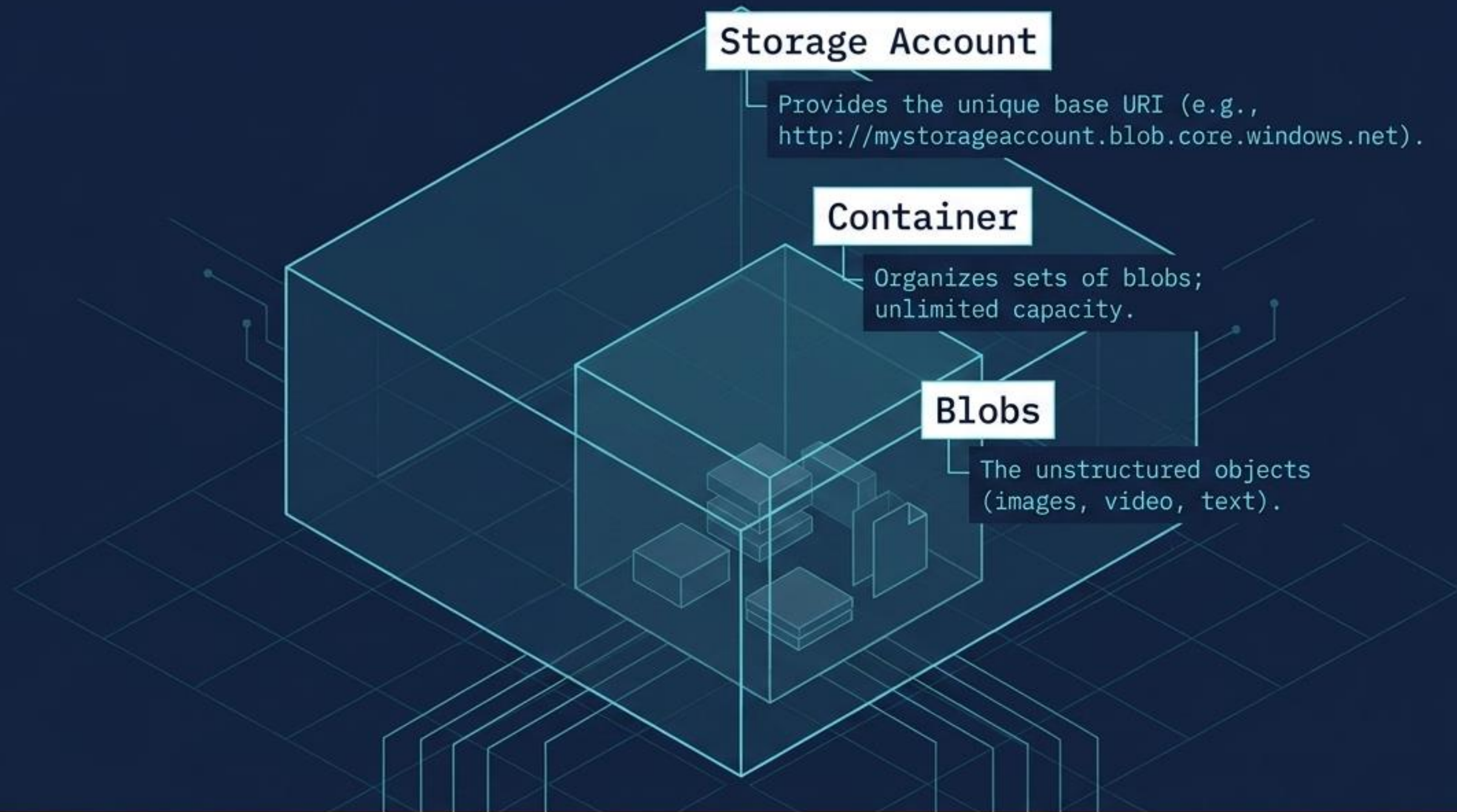


Architecting Data in Azure

A structural taxonomy of Microsoft's
cloud storage ecosystem

	Azure Blobs	Azure Files	Azure Disks	Azure Tables	Azure Queues
Data Type	Unstructured Objects	File Systems	Block-level Volumes	NoSQL Structured	Async Messages
Access Protocol	HTTP/REST	SMB / NFS	Native OS APIs	HTTP/REST	HTTP/REST
Primary Use Case	Massive data lakes & streaming	Lift-and-shift NAS replacement	Persistent VM storage	Schemaless flexible metadata	Application decoupling



Object storage functions as a massive key-value store over the internet, where the URI itself serves as the Key.



Block Blobs

- Function: Stores text and binary data managed as individual blocks.
- Scale: Massive; up to 190.7 TiB.
- Use Case: Standard object storage, serving images, document distribution.



Append Blobs

- Function: Block structure heavily optimized for append-only operations.
- Scale: Highly efficient for continuous writes.
- Use Case: Application logging and telemetry data.

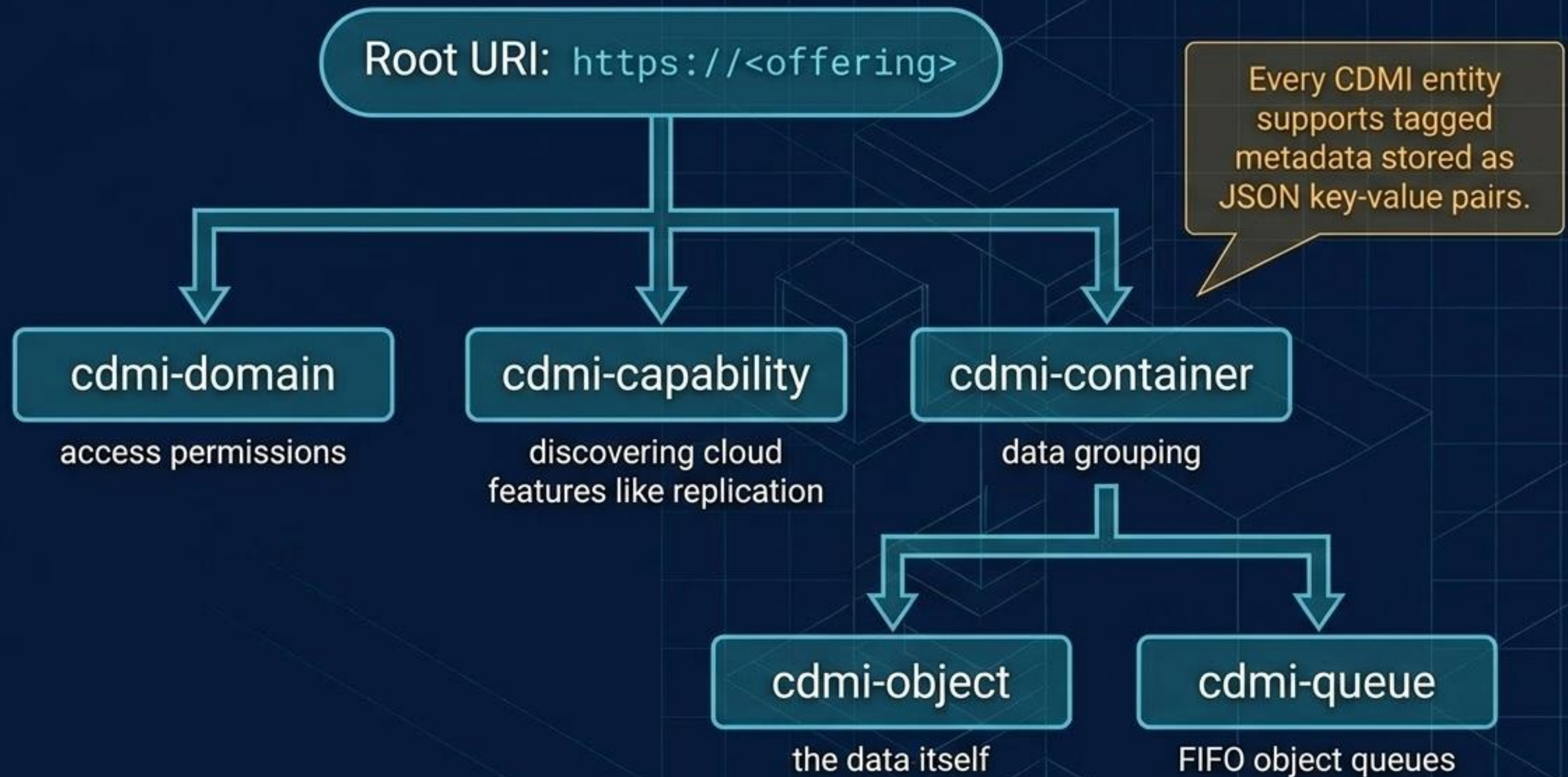


Page Blobs

- Function: Stores random-access files.
- Scale: Up to 8 TiB.
- Use Case: Acts as the underlying virtual hard drive (VHD) files for Azure VMs.



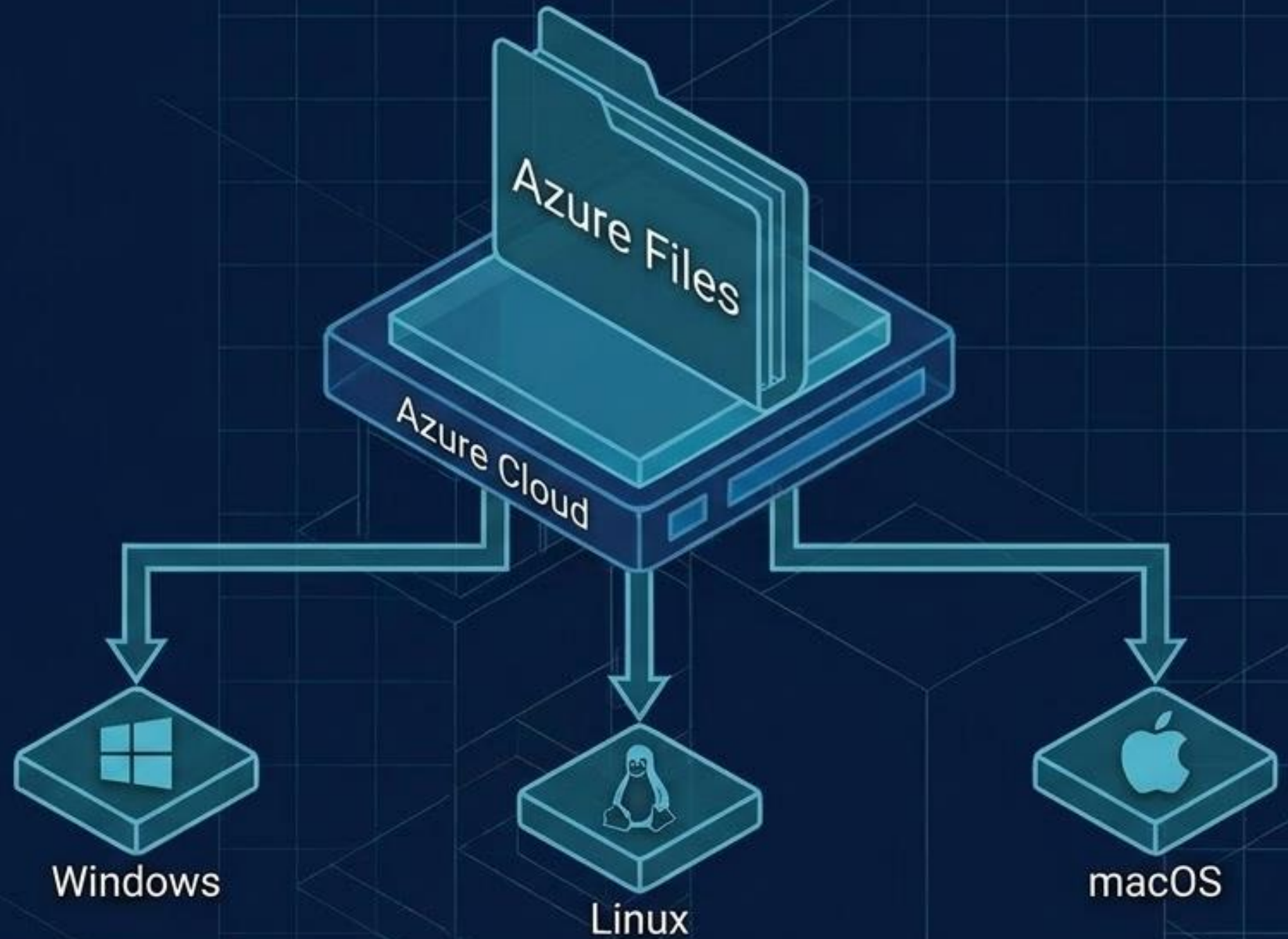
Azure acknowledges the request, validates the authorization, and confirms successful creation. No native update exists; objects are wholly overwritten.



Advantage: Enables seamless cloud peering and data migration without vendor lock-in.
Disadvantage: Industry adoption remains limited.

Core Mechanics

- Fully managed cloud file shares accessed via standard SMB or NFS protocols.
- Allows simultaneous read/write access from multiple virtual machines.



The Lift & Shift Advantage

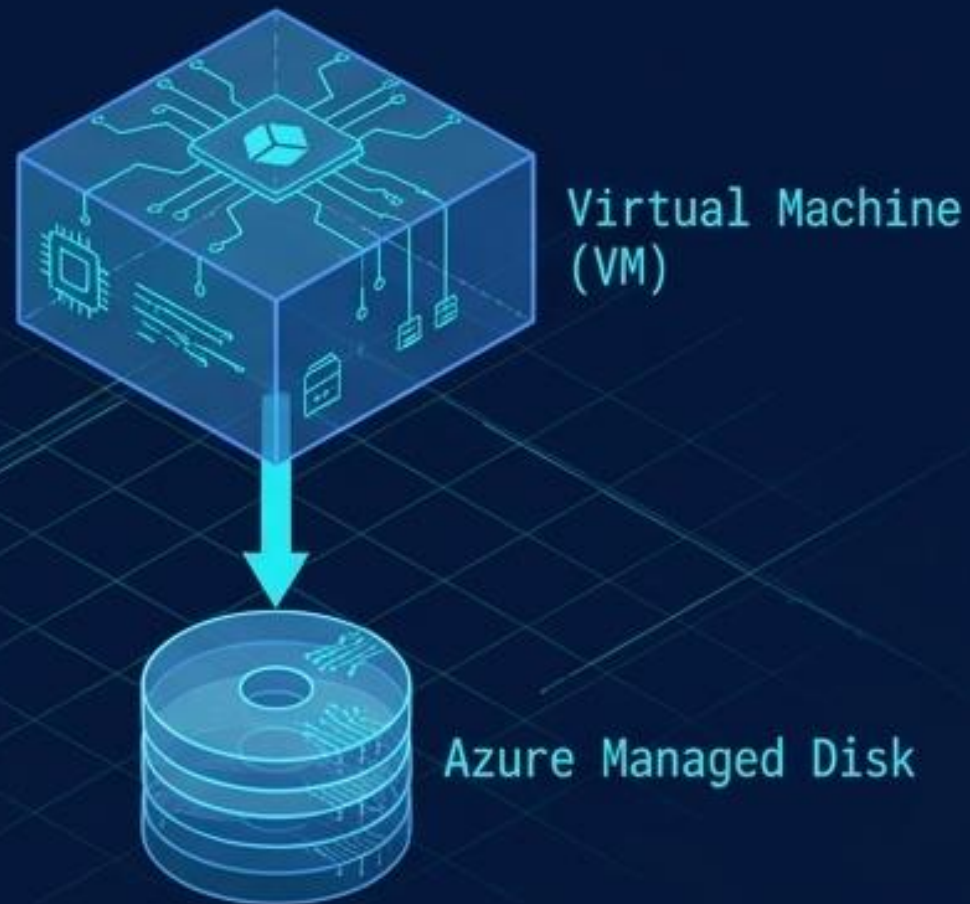
Migrate legacy applications instantly by mounting the Azure file share to the exact same drive letter the on-premises application already uses.



Security Callout: Access files globally via a URL secured by a Shared Access Signature (SAS) token, granting time-limited, specific permissions.

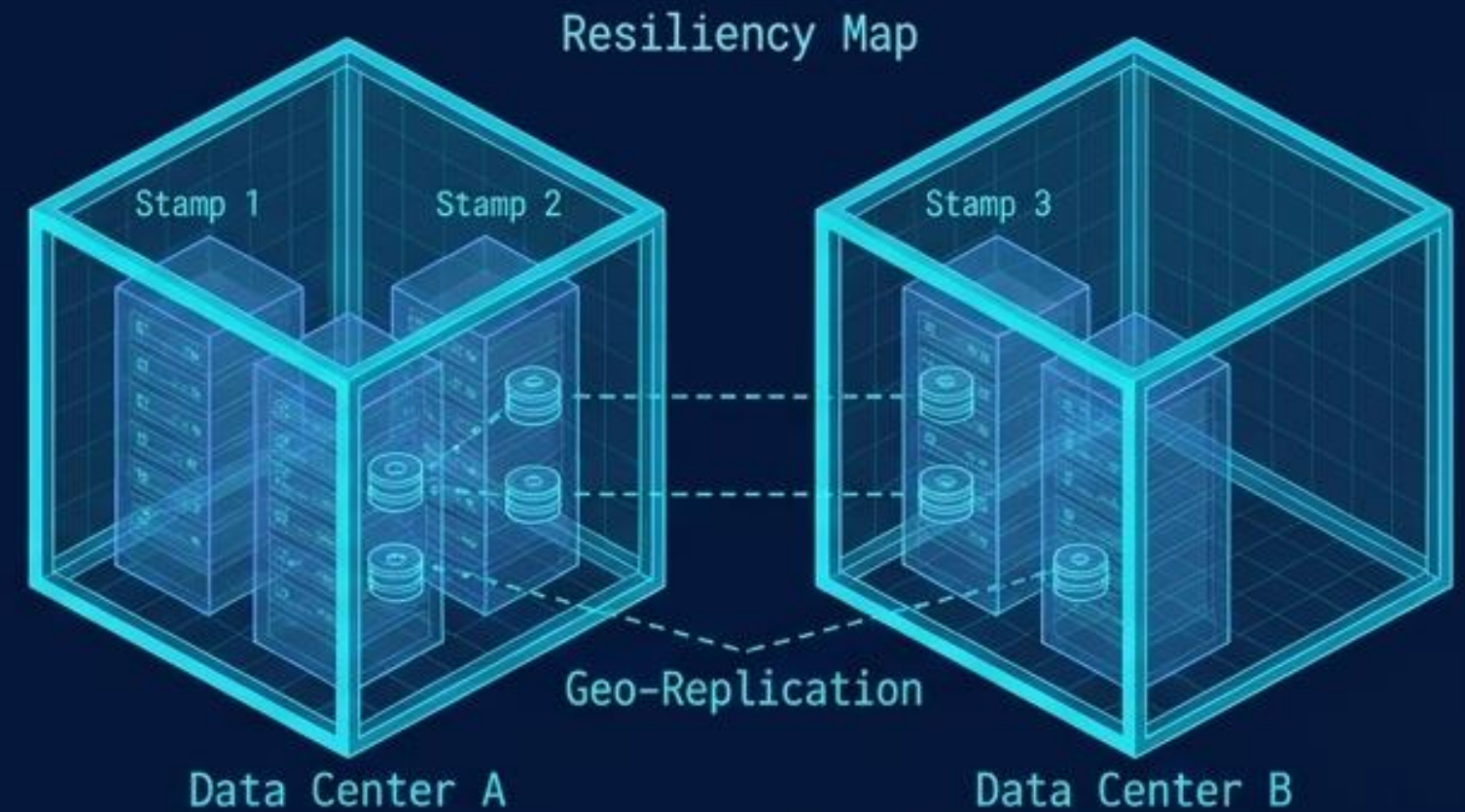
Azure Disk Storage: Function & Resiliency Blueprint

THE FUNCTION



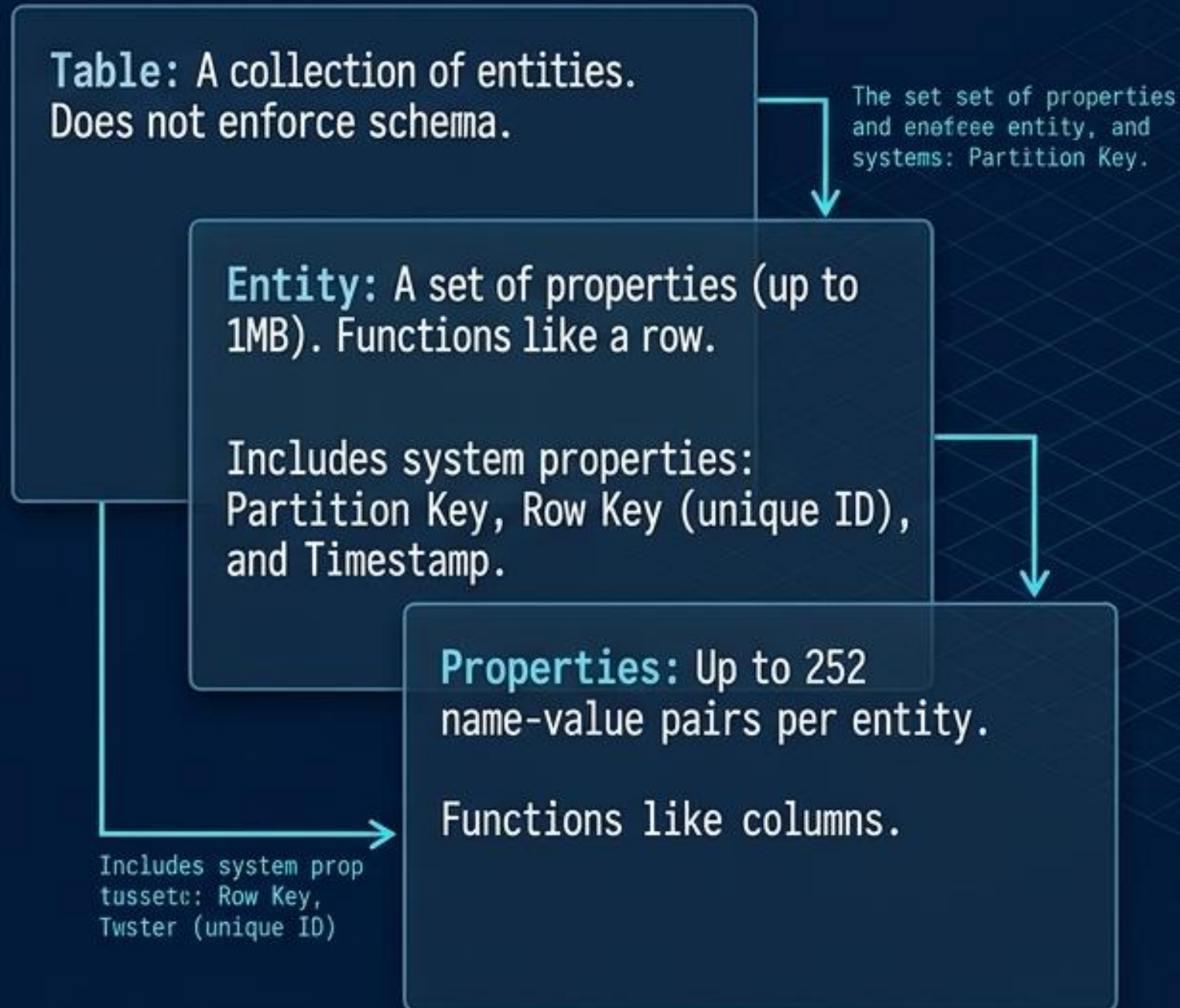
- Block-level storage volumes dedicated to Azure VMs.
- Data is persistently stored and accessed via native file system APIs.
- Not intended for access outside the attached VM.

THE RESILIENCY ARCHITECTURE

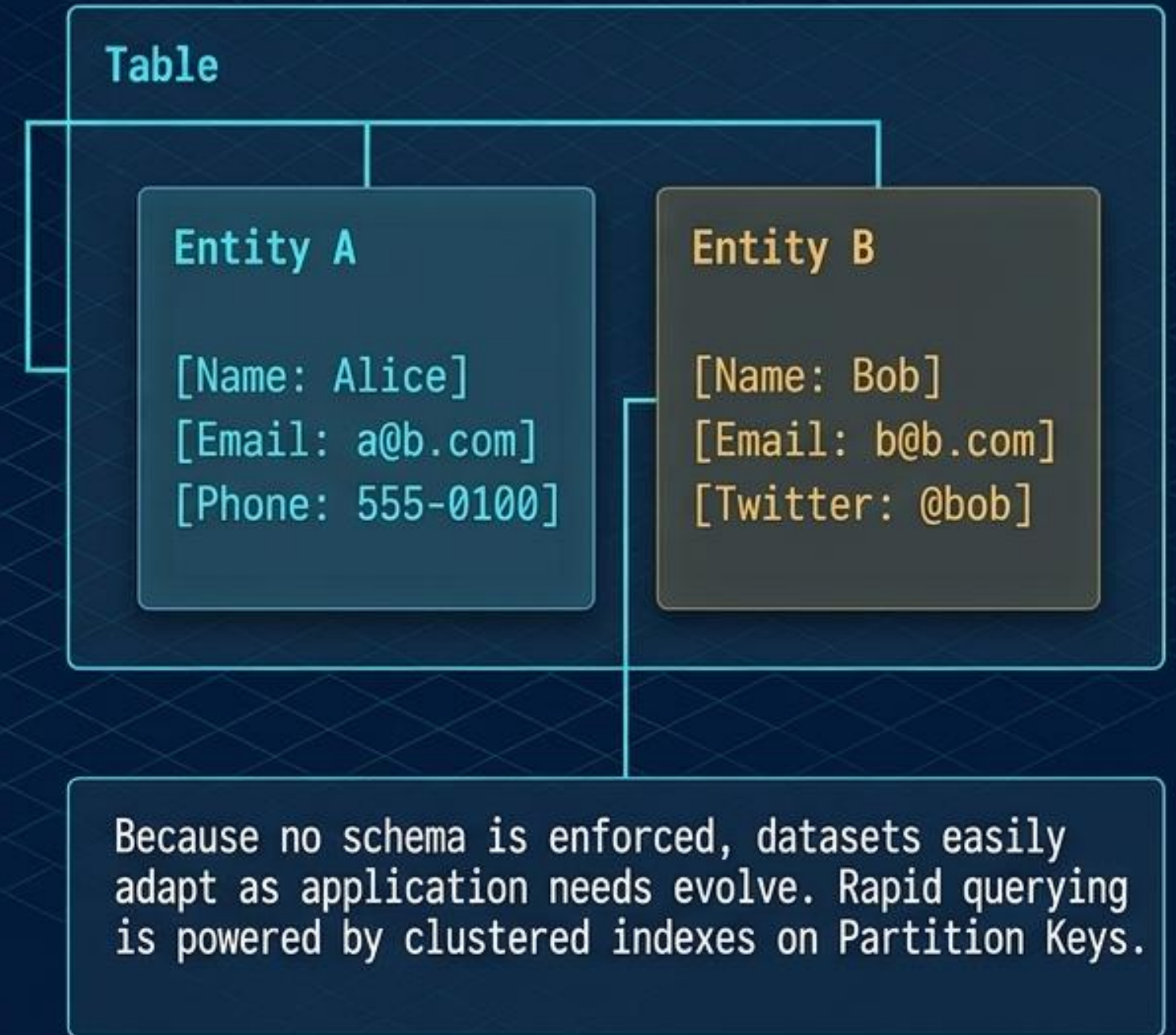


- Availability Sets: Disks are placed in different storage scale units (stamps). A hardware failure in one stamp only affects the disks directly on it.
- Availability Zones: Protects against total datacenter failure by isolating resources across unique physical locations with independent power, cooling, and networking.

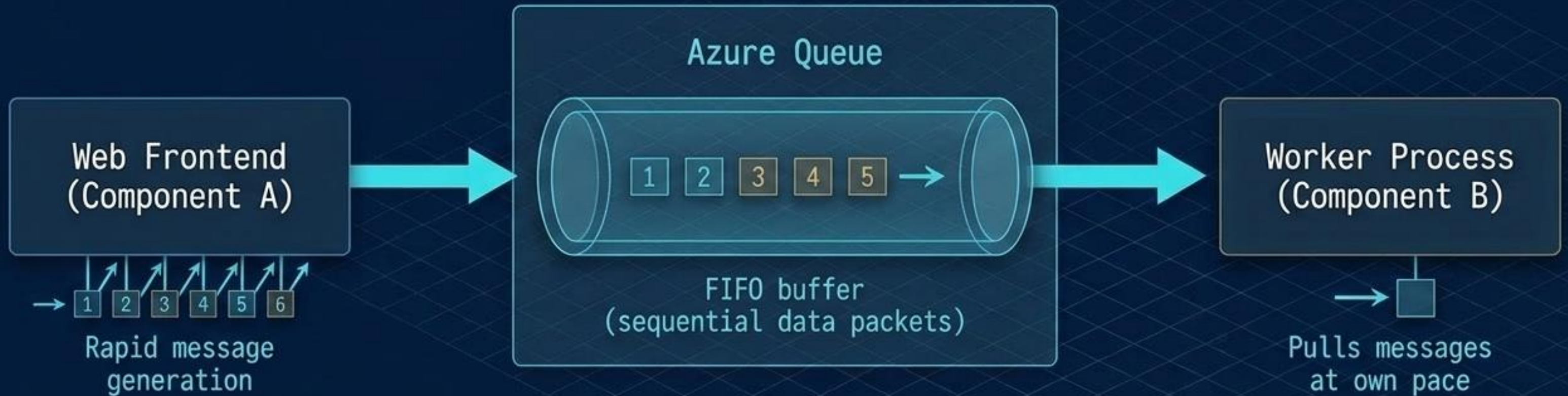
THE ANATOMY



THE "SCHEMALESS" EXAMPLE

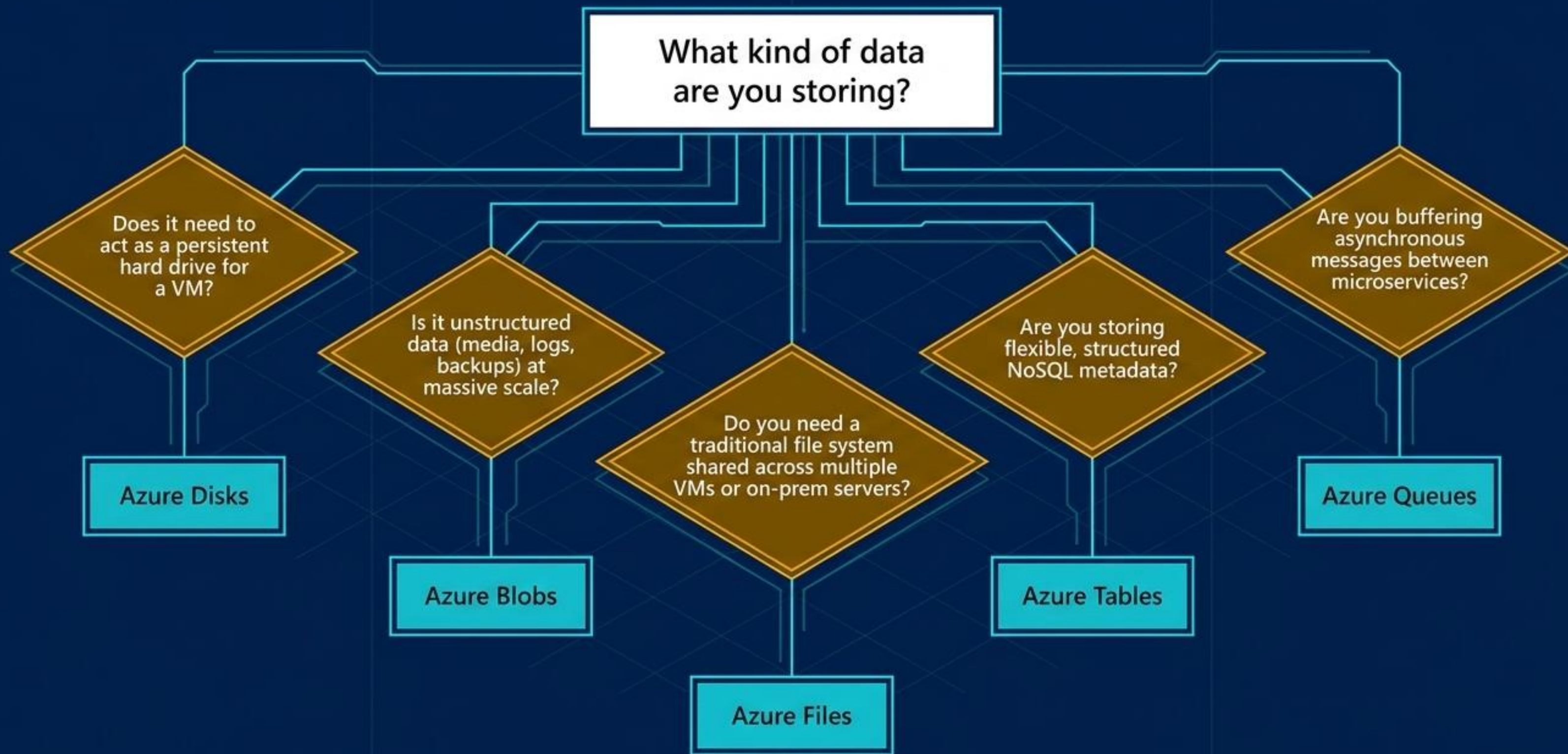


Asynchronous Message Queueing



Strategic Value

Decouples application components. If the worker process goes down or slows under heavy load, the frontend can continue operating normally while the queue safely buffers the requests.



Azure Storage Ecosystem: Predictable architectures for unpredictable scale.